



Development of powder bed fusion of pure copper with a 440 nm blue laser using wavelength beam combining of direct diode laser

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Abstract

A short-wavelength laser with a high power output and small spot size is required to improve the quality and efficiency of powder bed fusion (PBF) for pure copper parts. We employed a PBF with laser beam/metal (PBF-LB/M) system using a wavelength beam combining method to achieve a 440 nm blue laser with a maximum laser power of 400 W and spot size of 100 μm . We investigated the influence of laser power and scanning speed on the morphological changes of single tracks and evaluated the surface waviness of single layers fabricated at various overlap ratios. Based on these findings, a range of building conditions that reduce the porosity of the built parts was established. The results showed that the proposed PBF-LB/M system could produce copper parts on a pure copper substrate without preheating. A uniform geometry of a single track was obtained at a laser power of 300–350 W and scanning speed of 100–250 mm/s. For a single layer structure, the arithmetic mean height Pa of the built surface exhibited its highest value of 11 μm at an overlap ratio of 0.1, and decreased to 8.1 μm at an overlap ratio of 0.5. A relative density of 99.75% of the built structure was achieved at a volumetric energy density (VED) of 270 J/mm³. The porosity of the built structure decreased with an increase in the flow rate of the N₂ inert gas in the building chamber, and its value changed from 0.95% to 0.25% at a flow rate of 10–100 L/min. The high blue laser absorptivity of pure copper can reduce the VED by more than 50% compared with that of the infrared laser to obtain fully dense parts.

Keywords Powder bed fusion · Pure copper · Blue laser · Porosity · Gas flow rate

Abbreviations

AM additive manufacturing
PBF powder bed fusion
DED directed energy deposition
CS cold spraying

LB/M Laser beam/Metal
EB/M Electron beam/Metal
SEM scanning electron microscopy
VED Volumetric energy density

1 Introduction

Pure copper is an essential material with a thermal conductivity of 397 W/m²K and electrical conductivity of 102% International Annealed Copper Standard (IACS) [1], and its properties are advantageous for aerospace components, electronics, and thermal management systems [2]. In recent years, additive manufacturing (AM) techniques such as powder bed fusion (PBF), directed energy deposition (DED), and cold spraying (CS) have been applied in the manufacturing of copper parts to achieve complex features with acceptable mechanical properties [3]. Qu et al. [4] developed a high precision PBF process using a fibre laser

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with a wavelength of 1070 nm, small laser spot of 25 μm , fine powder diameter of 5–25 μm , and thin layer thickness of 10 μm to obtain dense copper parts at a low-power output and revealed that a relative density of 99.6% and electrical conductivity of 76.1% IACS for the built part was achieved at a laser power of 200 W, scanning speed of 600 mm/s, and hatching distance of 0.05 mm. Huang et al. [5] compared the electrical conductivity of copper parts fabricated via CS and PBF using a Yb-fibre laser with a wavelength of 1064 nm and spot size of 100 μm and found that the electrical conductivity of the PBF-fabricated part was lower than that of the CS-fabricated part due to the partially molten powders and voids in the built parts. Yadav et al. [6] established a process parameter window for DED using a fibre laser with a high laser power of 2 kW, and revealed that a laser energy per unit powder supply of 22.5 kJ/g offered complete melting and diffusion between the adjacent layers, leading to a relative density of 99.9%. In the CS process, the built parts suffer from low density, and further post-processing, such as hot isostatic pressing, is required to obtain dense parts. Meanwhile, the PBF and DED processes using an infrared laser require a high-power output to achieve dense parts. These studies revealed that it is imperative to select an appropriate AM technique to improve the quality of copper parts because porosity and defects affect their thermal, electrical, and mechanical properties.

In the AM processes using a conventional infrared laser with a wavelength of approximately 1 μm , the required energy was found to be considerably higher than that of lasers with short wavelengths because the laser absorption rate was less than 10% [7]. In addition to its low laser absorptivity, the high thermal conductivity of copper causes rapid melting/solidification of the molten metal, resulting in cracking [8], a nonuniform distribution of precipitates at the boundary [9], and partially molten powders and voids in the built part [5]. To suppress these defects, AM using short-wavelength lasers with high copper absorption rates has attracted attention [10]. Hori et al. [11] developed a PBF system using a blue laser with a wavelength of 445 nm and investigated the influence of scanning speed and hatching distance on the density of copper parts with dimensions of $5 \times 5 \times 5 \text{ mm}^3$. At a laser power of 200 W, scanning speed of 40 mm/s, and hatching distance of 50 μm , a built part with a density of approximately 99.1% was obtained. Ono et al. [12] evaluated the electrical resistivity of copper parts fabricated via DED using a blue laser with a wavelength of 450 nm and showed that the electrical resistivity of the built part fabricated at a laser power of 30 W was similar to that of commercial copper, whereas the built part fabricated at 60 W exhibited a higher electrical resistivity owing to oxidation by the residual heat of the deposited layer. Asano et al. [13] investigated the effect of

laser wavelengths of 450 and 915 nm on the building characteristics of copper parts via DED and revealed that a blue laser required lower laser intensity to obtain a stable single track. Copper parts without voids were obtained at laser intensities of 69 and 108 kW/cm² for the blue and infrared lasers, respectively. Liu et al. [14] focused on the DED process of copper using a blue laser with a wavelength of 450 nm and found that the density of built parts increased with increasing laser power. A density of 99.6% was achieved at a laser power of 650 W. Takenaka et al. [15] fabricated copper parts on different substrate materials via DED using a blue laser with a wavelength of 450 nm and showed that the aluminium substrate was not melted using only laser irradiation; the deposited layer was formed by melting aluminium based on heat conduction from molten copper powder. These results indicate that the absorption characteristics of the blue laser are effective in stabilising the building process of copper parts.

Meanwhile, thermal management systems, such as heat exchangers and heat pipes, require functional surfaces with large surface areas to increase thermal efficiency. These systems are essential for safe and efficient performance in a variety of engineering applications such as lithium-ion batteries, electric vehicles, and renewable energy. When AM techniques are applied to manufacture these parts, the PBF process is critical for fabricating fine parts with complex features because of its high resolution. Generally, the PBF process utilises laser beam/metal (PBF-LB/M) and electron beam/metal (PBF-EB/M) as heat sources. The vacuum environment in PBF-EB/M is advantageous for fabricating aluminium and titanium alloys that are sensitive to the capture of interstitial impurities such as oxygen [16]. However, it is difficult to achieve high resolution owing to the sticking of metal powders during raking [17]. In contrast, the PBF-LB/M process offers a variety of combinations of energy density, laser wavelength, powder diameter, and layer thickness that allow exploring the possibilities of high-precision manufacturing. Ma et al. [18] investigated the influence of process parameter and scanning strategy on the geometry, surface roughness, and thickness of thin-walled parts via PBF using a fibre laser with a wavelength of 1070 nm and showed that a relative density of 99.88%, wall thickness of 143 μm , and surface waviness of 13 μm were achieved at a laser power of 500 W, scanning speed of 1000 mm/s, and hatching distance of 60 μm using a circular scanning strategy. Horn et al. [19] fabricated copper parts via PBF using a green disk laser with a wavelength of 515 nm and found that a thin-walled part with a minimum core thickness of 230 μm was obtained using a green laser with a beam diameter of 200 μm . In contrast to infrared lasers, short-wavelength lasers appear to have a minimum spot size of 200 μm [11, 19], and the reviewed literature

suggests that the current hardware capabilities of short-wavelength lasers are not as refined as those of infrared lasers for creating a high-resolution geometry with smooth surfaces [18]. A short-wavelength laser with a high power output and small spot size is essential for improving the quality and efficiency of PBF for pure copper parts.

Many studies on PBF-LB/M of pure copper have utilised stainless steel as a substrate material, such as AISI 304 [12–15] and 316 L [4, 5], because of its little dependence of absorptivity on the laser wavelength [20]. Owing to the low thermal conductivity of steel, the use of steel substrates instead of copper substrates has become a common approach to prevent rapid heat loss from the melt pool during building, which can lead to delamination of the initial layer [2]. In a hybrid process, such as a direct AM technique for metal parts [21], the building characteristics of pure copper powder on a pure copper substrate are important, but not well understood. Moreover, studies on the PBF of pure copper using blue lasers are lacking. Therefore, we adopted a PBF-LB/M system using a wavelength synthesis method with a direct diode laser (DDL). To obtain a dense part on the pure copper substrate, a blue laser with a wavelength of 440 nm, maximum laser power of 400 W, and spot size of 100 μm was utilised. The aims of this study were threefold: to investigate the influence of laser power and scanning speed on the morphological change of a single track, to evaluate the surface waviness of a single layer fabricated at various overlap ratios, and to establish a range of building conditions that reduce the porosity of the built part. Based on these results, single and multilayer formations in the PBF process with a 440 nm blue laser were systematically examined.

2 Methodology

The experimental setup is shown in Fig. 1, and the building conditions are listed in Table 1. The PBF-LB/M prototype machine (Panasonic Connect Co. Ltd.) employs the wavelength beam combining method using a DDL [22]. To achieve a high-power output of 400 W with a wavelength of 440 nm, laser beams with different wavelengths emitted from the DDL were focused on a diffraction grating, and then the focused beam through a core fibre with a diameter of 50 μm was irradiated using a galvanometer scanner. The profile of the focused beam with a Gaussian distribution was measured using a beam profiler (BPF Laser Innovation Corp., BPF-S400), and the $1/e^2$ width of the beam was defined as the spot diameter. The spot diameter was adjusted to 102 μm , and the focal position of the beam was located at the surface of a powder bed. A scanning electron microscopy (SEM) image of the metal powder is shown in Fig. 2. The metal powder was pure copper with an average diameter



Fig. 1 Experimental setup

Table 1 Building conditions

Metal powder		Pure copper
Mean diameter	[μm]	44
Substrate		Pure copper
Size	[mm]	70 × 70 × 10.5
Laser irradiation		
Laser type		Semiconductor
Wavelength	[nm]	440 ± 20
Laser power P	[W]	100–350
Scanning speed v	[mm/s]	50–300
Hatching distance H	[μm]	80, 112, 144
Spot size ϕ	[μm]	102
Layer thickness t	[μm]	80
Shielding gas		N_2
Flow rate	[L/min]	10–100

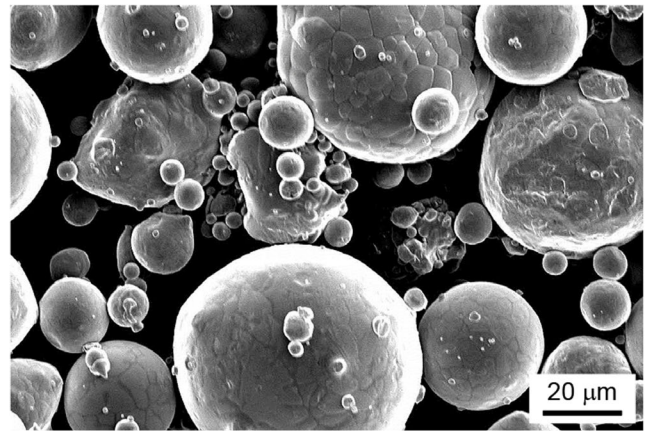


Fig. 2 SEM image of copper powder

(D_{50}) of 44 μm . To evaluate the powder properties, the angle of repose and bulk density were measured according to JIS Z 2502 and JIS Z 2504, respectively. The properties of the powders are listed in Table 2. The pure copper substrate had

Table 2 Powder properties

Density (kg/m ³)	4,590
Circularity (%)	65
Angle of repose (°)	28

dimensions of 70 mm × 70 mm × 10.5 mm. The substrate was sandblasted with alumina abrasive grains to improve the wettability of the molten powder. The building chamber was filled with N₂ inert gas, and the building pitch in the vertical direction was set to 80 μm.

First, the influence of process parameters such as laser power and scanning speed on the morphological change of a single track was investigated. A single track with a length of 20 mm was built on a pure copper substrate at laser powers of 100–350 W and scanning speeds of 50–300 mm/s. The morphological changes in the single track were evaluated using an optical microscope (Keyence Corp., VHX-S15H), and the process window for deducing the distortion of the single track was established. Based on the results in the single track formation, the building condition without distortion was applied to the formation of a single layer, and 100 line tracks were built at an overlap ratio of 0.1–0.5 relative to the width of a single track. The overlap ratio was determined using the following equation:

$$\text{Overlap ratio} = 1 - \frac{H}{w} \quad (1)$$

where H is the hatching distance and w is the built width.

To evaluate waviness, the surface profile of a single layer was measured using a surface roughness tester (Tokyo Seimitsu Co. Ltd., SURFCOM 2000DX2), and the arithmetic mean height of the primary profile P_a [23] was evaluated. Thereafter, the building process was repeated for 40 layers using a zigzag scanning strategy, and the scanning direction was rotated 90° for each layer to obtain a multilayer structure. To maintain a constant thickness of the powder bed and determine the building pitch in the vertical direction, the cross-sectional profile of the built structure in each layer was evaluated using a three-dimensional measuring system (Keyence Corp., VR-5000). The built height of each layer was evaluated from the obtained cross-sectional profiles, and the built height per layer was calculated by subtracting the built height of the previous layer from that of the subsequent layer. The building experiment proceeded by adjusting the lowering amount of the building platform during the squeezing process based on the measured built height per layer, and the quality of the built structure was evaluated. As the bulk density of the powder (4,590 kg/m³) is approximately half that of pure copper (8,960 kg/m³), the cumulative lowering amount of the building platform ΔZ [μm], the thickness of the powder-bed layer t [μm], and the

cumulative built height h [μm] can be calculated using the following equations, under the assumption that all the powder is fully melted and contributes to the formation of the built structure, resulting in a halving of the built volume.

$$\Delta Z_n = P_v \times n \quad (2)$$

$$t_n = \Delta Z_n - h_{n-1} \quad (3)$$

$$h_n = h_{n-1} + \frac{t_n}{2} \quad (4)$$

where P_v is the building pitch in the vertical direction and n is the number of layers. Moreover, the predicted built height per layer can be obtained by subtracting the cumulative built height of the previous layer from that of the subsequent layer.

For the cross-sectional observation, the built structure was sectioned using a precision cutting machine (Nippura Corp., CUT-242) and polished using a polishing machine (Refine Tech Co., Ltd., RP0-128B). To evaluate the porosity of the built structure, an optical micrograph was binarised using an open-source software (ImageJ). Additionally, the influences of laser wavelength and substrate material on the porosity were examined using the volumetric energy density (VED) [J/mm³], representing the energy density per unit volume of the material supply, as determined in Eq. (1) [24]:

$$VED = \frac{P}{V \cdot H \cdot t} \quad (5)$$

where P is the laser power, v is the scanning speed, H is the hatching distance, and t is the thickness of the powder-bed layer.

Furthermore, to investigate the influence of the N₂ inert gas flow rate in the building chamber on the quality of the built structure, the porosities of structures fabricated at flow rates of 10, 20, 40, and 100 L/min were assessed.

3 Experimental results and discussion

3.1 Influence of laser power and scanning speed on single track formation

The morphological changes of single tracks at various laser powers are shown in Fig. 3. At a laser power of 250 W, the distortion of the single track became significant, as shown in Fig. 3(a). The geometry of the single track was improved as the laser power increased (Fig. 3(b)), and a uniform semicircular geometry with a width of 160 μm was obtained at a laser power of 350 W (Fig. 3(c)). These results indicate that a higher laser power can provide a more stable process.

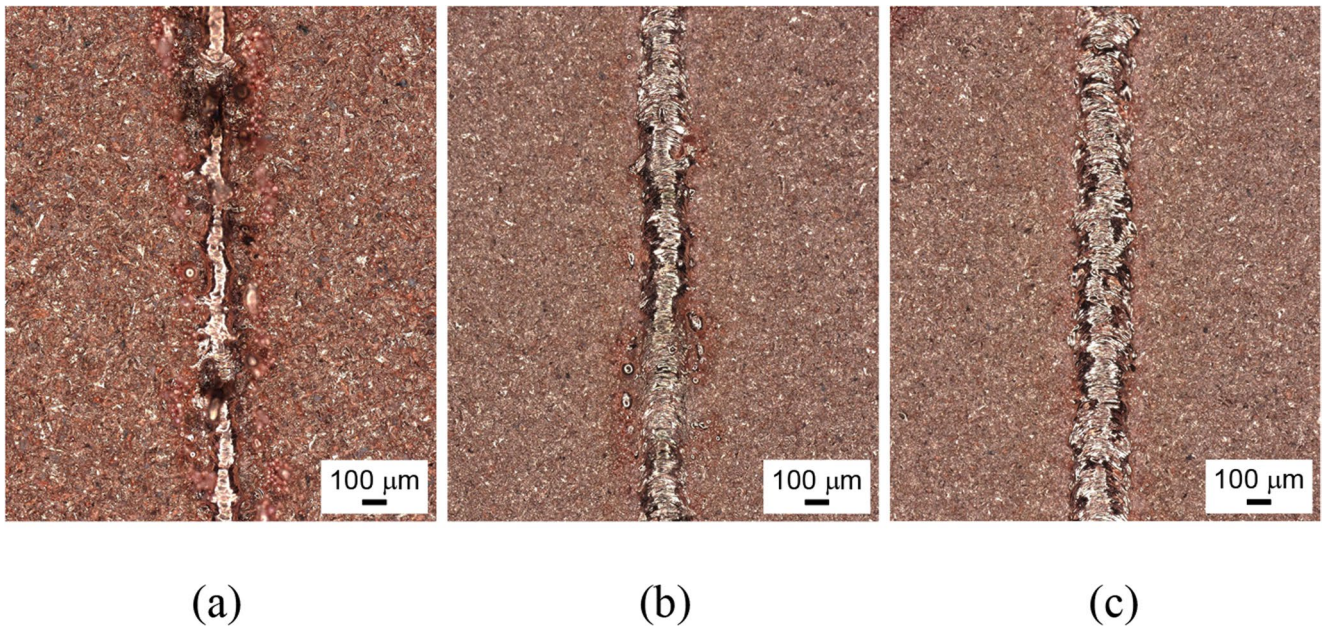


Fig. 3 Morphological change of single tracks at laser powers of (a) 250 W, (b) 300 W, and (c) 350 W (scanning speed: 200 mm/s)

In this study, both the powder and substrate are pure copper; therefore, the high reflectivity and high thermal conductivity of pure copper are considered to influence process stability. During the PBF-LB/M process, the laser beam transfers energy to the powder surface through reflection, absorption, and scattering. The unabsorbed laser beam passes through the powder and is absorbed by the substrate surface. Owing to its lower absorptivity, pure copper powder may allow a larger portion of the laser beam to be absorbed at the substrate surface compared with more absorptive powders. Mosavi et al. [25] reported that the temperature at the topmost surface was lower than that at the nadir surface immediately after laser irradiation due to the energy released associated with the formation of the molten metal and the heat loss caused by heat conduction through the substrate. Accordingly, the high thermal conductivity of pure copper promotes heat loss to the substrate, which significantly reduces the temperature at lower laser power and causes distortion of the single track. These results indicate that heat loss due to thermal conduction into the pure copper substrate is a critical factor affecting the process stability.

The morphological changes of single tracks at various scanning speeds are shown in Fig. 4. At a scanning speed of 50 mm/s, the single track was distorted owing to instability in the building process, as shown in Fig. 4(a). By contrast, a uniform single track with a width of 93 µm was observed at a scanning speed of 250 mm/s, as shown in Fig. 4(b). When a stainless steel substrate was used, stable built structures were obtained at lower scanning speeds [11], whereas the opposite trend was observed for a pure copper substrate. The high reflectivity and high thermal conductivity of

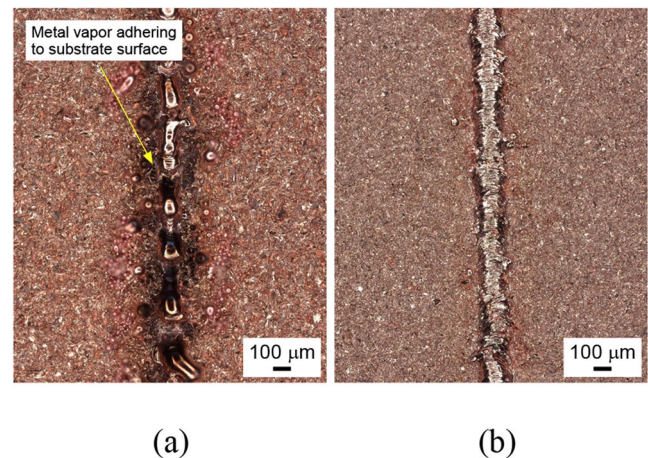


Fig. 4 Morphological change of single tracks at scanning speeds of (a) 50 mm/s and (b) 250 mm/s (laser power: 300 W)

pure copper can result in excessive material evaporation and keyhole instability. Blom et al. [26] reported that the absorptivity gradually increased with temperature, but sharply increased near the phase transition from solid to liquid. This can generate an instantaneous increase in the heat input, which leads to sudden evaporation of the melt pool and results in spatter generation. Moreover, this effect rises the recoil pressure acting on the melt pool surface, leading to a deeper keyhole and a stronger convection in the melt pool. Strong Marangoni convection, driven by surface tension gradients under high temperature gradients, promotes spatter generation. Moreover, the intrusion of metal vapour can lead to deterioration of the beam quality [27], thereby promoting an unstable process. As shown in

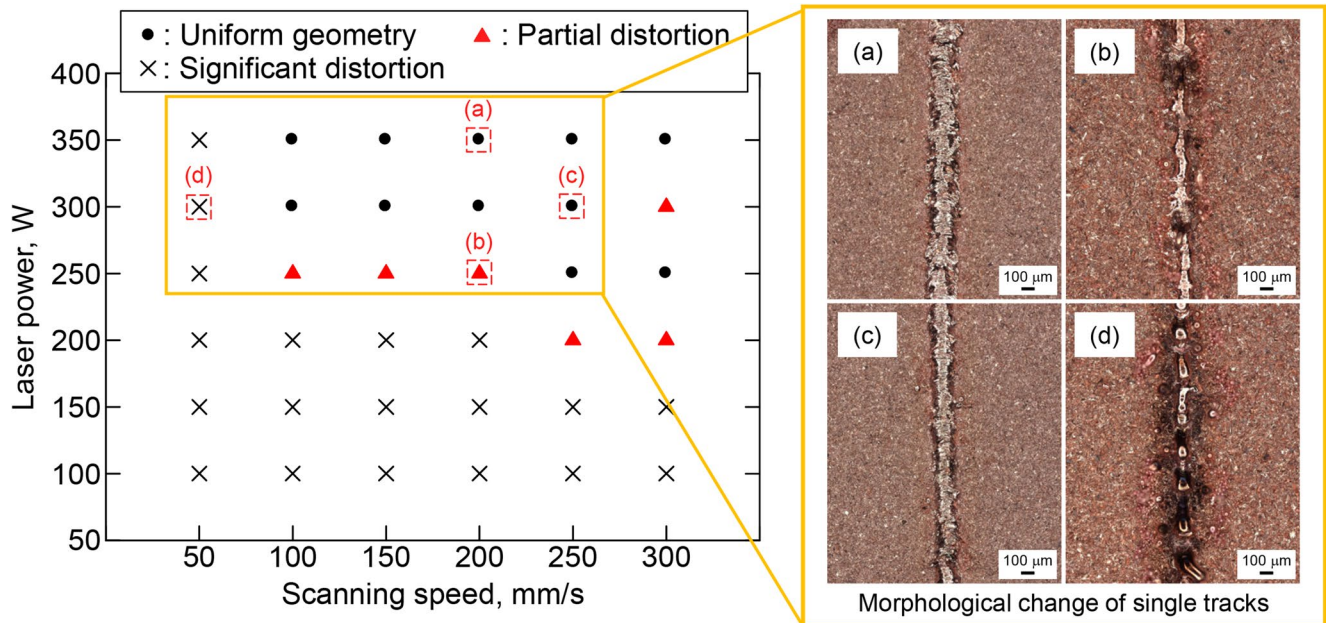


Fig. 5 Process window for obtaining good quality of a single track

Fig. 4(a), the metal vapour adhered to the substrate surface. Notably, the single track formed on the substrate in Fig. 4(a) appears smaller than that in Fig. 4(b), despite using identical processing conditions except for the scanning speed. This difference can be attributed to spatter loss. In summary, the distortion of a single track can be caused by interactions among these factors at an excessively low scanning speed.

The process window for obtaining a good-quality single track is presented in Fig. 5. As illustrated in Figs. 3 and 4, the morphologies of single tracks were classified as uniform geometry (○), partial distortion (△), and significant distortion (×). At a laser power of 100–150 W and a scanning speed of 300 mm/s, discontinuous structures were formed owing to a lack of heat input, whereas the process was unstable when the scanning speed was 50 mm/s. As a result, uniform structures were obtained at a laser power of 300–350 W and scanning speed of 100–250 mm/s.

3.2 Influence of the overlap ratio on the single layer formation

In this study, the influence of the overlap ratio on the surface waviness of a single layer was investigated. The variation in the arithmetic mean height P_a of single layers fabricated at overlap ratios of 0.1, 0.3, and 0.5 is depicted in Fig. 6, and the surface profiles of the single layers are shown in Fig. 7. For single-layer formation, a laser power of 350 W and scanning speed of 200 mm/s were selected because the distortion of the single track was minimal. As the width of the single track was 160 μm, the overlap ratios of 0.1, 0.3, and 0.5 represent hatching distances of 144, 112, and 80 μm, respectively. The

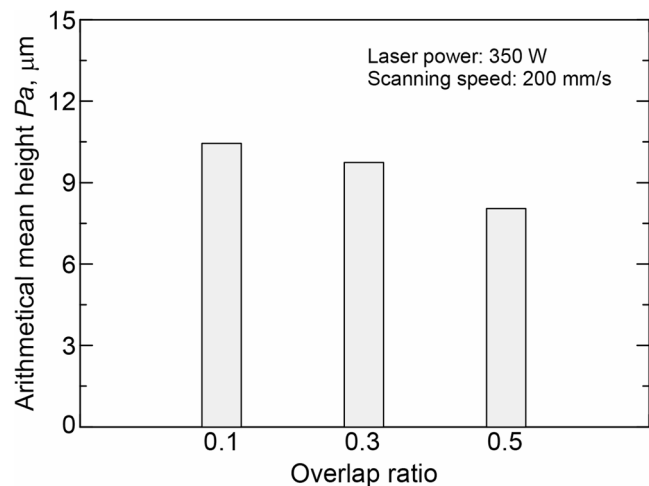


Fig. 6 Variation in the arithmetic mean height of single layers at different overlap ratios

arithmetic mean height of the single layer reduced with an increase in the overlap ratio and its value was 11 μm or lower. The arithmetic mean height was the smallest, 8.1 μm, at an overlap ratio of 0.5. At a smaller overlap ratio, the curvature between adjacent tracks increased, resulting in the deterioration of the surface quality of the built structure. Additionally, a large area of adjacent tracks was remelted during laser irradiation when the overlap ratio was large, which may have improved the surface quality of the built structure. For this effect, Grub et al. [28] investigated the laser polishing of PBF-fabricated copper parts using a visible laser wavelength of 515 nm and revealed that the surface roughness was reduced from 21.6 to 3.2 μm R_a due to re-melting of the built surface.

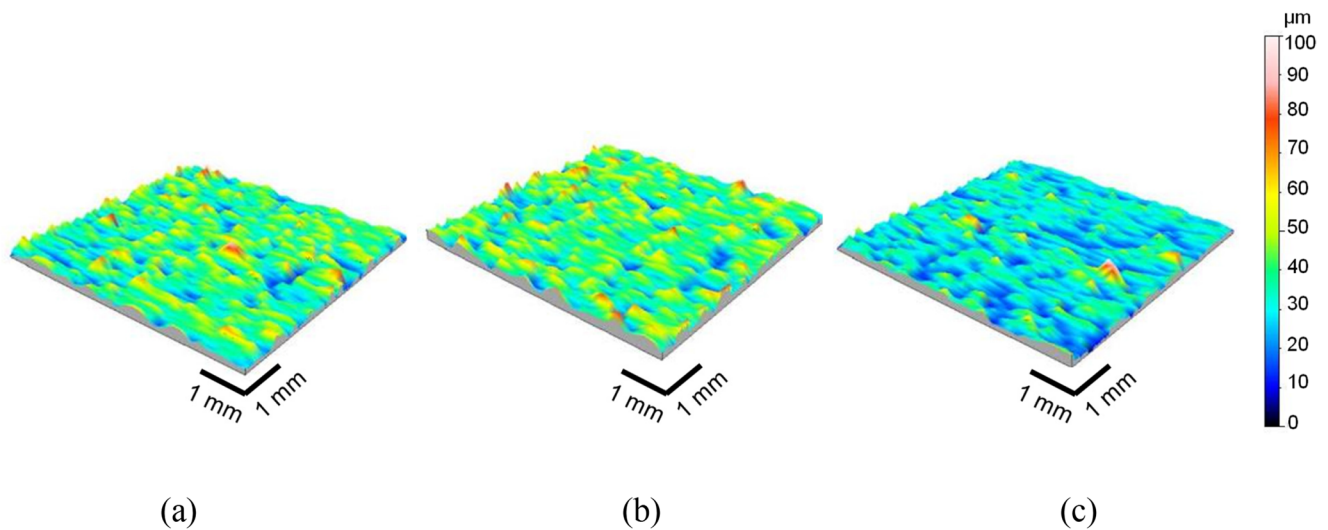


Fig. 7 Surface profiles of single layers at overlap ratios of (a) 0.1, (b) 0.3, and (c) 0.5

Based on these results, it is imperative to select an overlap ratio based on the geometry of a single track to obtain a built structure without surface waviness.

3.3 Assessment of the porosity in the multilayer structure

A cross-sectional image of the built structure fabricated at a laser power of 350 W, scanning speed of 200 mm/s, and overlap ratio of 0.5 is displayed in Fig. 8. When the built structure was divided into three locations in the height direction, the porosity increased with an increase in the built height, and the value of the entire cross section was 1.6%. To investigate the factors that affect the porosity, the variation in the built height in each layer was evaluated.

A comparison of the predicted built height ΔZ , measured built height, and predicted thickness of the powder-bed layer t is shown in Fig. 9. When all the powder is fully melted and contributes to the formation of the built structure, the built volume is reduced by half because the bulk density of the powder (4,590 kg/m³) is approximately half that of pure copper (8,960 kg/m³). The powder layer thickness

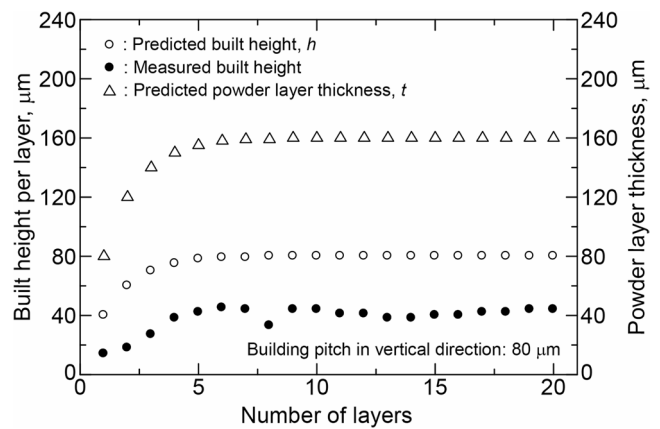
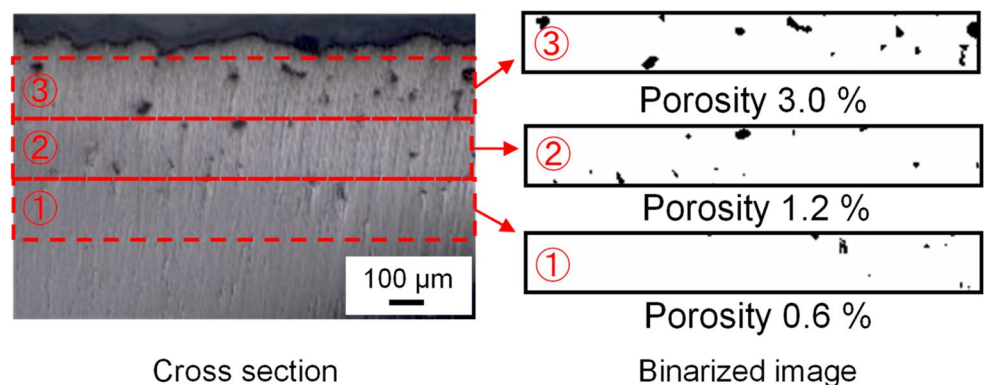


Fig. 9 Comparison of the predicted built height h , measured built height, and predicted powder layer thickness t (building pitch in vertical direction: 80 μm)

t_1 and built height h_1 in the first layer were calculated to be 80 and 40 μm, respectively. In the second layer, t_2 and h_2 were calculated as 120 and 100 μm, whereas t_3 and h_3 were calculated as 140 and 170 μm in the third layer, corresponding to the built heights per layer of 60 and 70 μm,

Fig. 8 Variation in porosity of the built structure in the height direction



respectively. The powder layer thickness varied with the number of layers because of the difference in the density of the powder bed and built structure, and their values stabilised at approximately $160\text{ }\mu\text{m}$ after the eighth layer. The measured built heights showed trends similar to the predicted ones, but the absolute values differed significantly. Moreover, the measured built height was approximately 30% of the powder layer thickness. Under unstable building conditions, the presence of porosity and voids caused by lack-of-fusion in the built structure can affect the relative density. Hyer et al. [29] revealed that porosity due to a lack of fusion was observed at a thicker powder layer, and the relative density decreased to 60% under inappropriate layer thickness. In addition, Malý et al. [30] reported that a relative density over 99% was achieved at a thin layer thickness of $30\text{ }\mu\text{m}$ and a high preheating temperature of $400\text{ }^{\circ}\text{C}$. When the porosity and voids remain inside the built structure, the apparent volume of the structure increases. Therefore, they are not considered to be the dominant factors contributing to the reduction in the built height. Another factor that may contribute to material loss is the spatter generation from the laser irradiation area. Kaufmann et al. [31] showed that the number of spatter particles with a wide size range increased, leading to a greater mass loss. In addition, Fujio et al. [32]

showed that the blue laser can reduce the sudden increase in the heat input owing to a smaller change in absorptivity during the phase transition compared with the infrared laser, resulting in a stable process and low spattering. The low height of the built parts can be caused by the interactions between an excessive thickness of the powder layer and the mass loss owing to spatter generation, although visible lasers can suppress the process instability caused by the difference in absorptivity during the phase transition. In addition, the measured built height became nearly constant as the number of layers increased. This may be attributed to heat accumulation during the building process and to the limitation of the heat diffusion region caused by thermal conduction as the built height increased. These results indicate that a combination of a high-power laser, thin powder layer, and preheating is required to obtain dense copper parts, and that an excessive thickness of the powder layer causes excessive spatter loss and porosity in the built parts.

Based on the results shown in Fig. 9, the height of the obtained built structure was approximately 30% of the powder layer thickness. Therefore, the lowering amount of the building platform (i.e., the building pitch in the vertical direction) in each layer was adjusted to maintain a constant thickness of the powder bed at $80\text{ }\mu\text{m}$, as shown in Fig. 10.

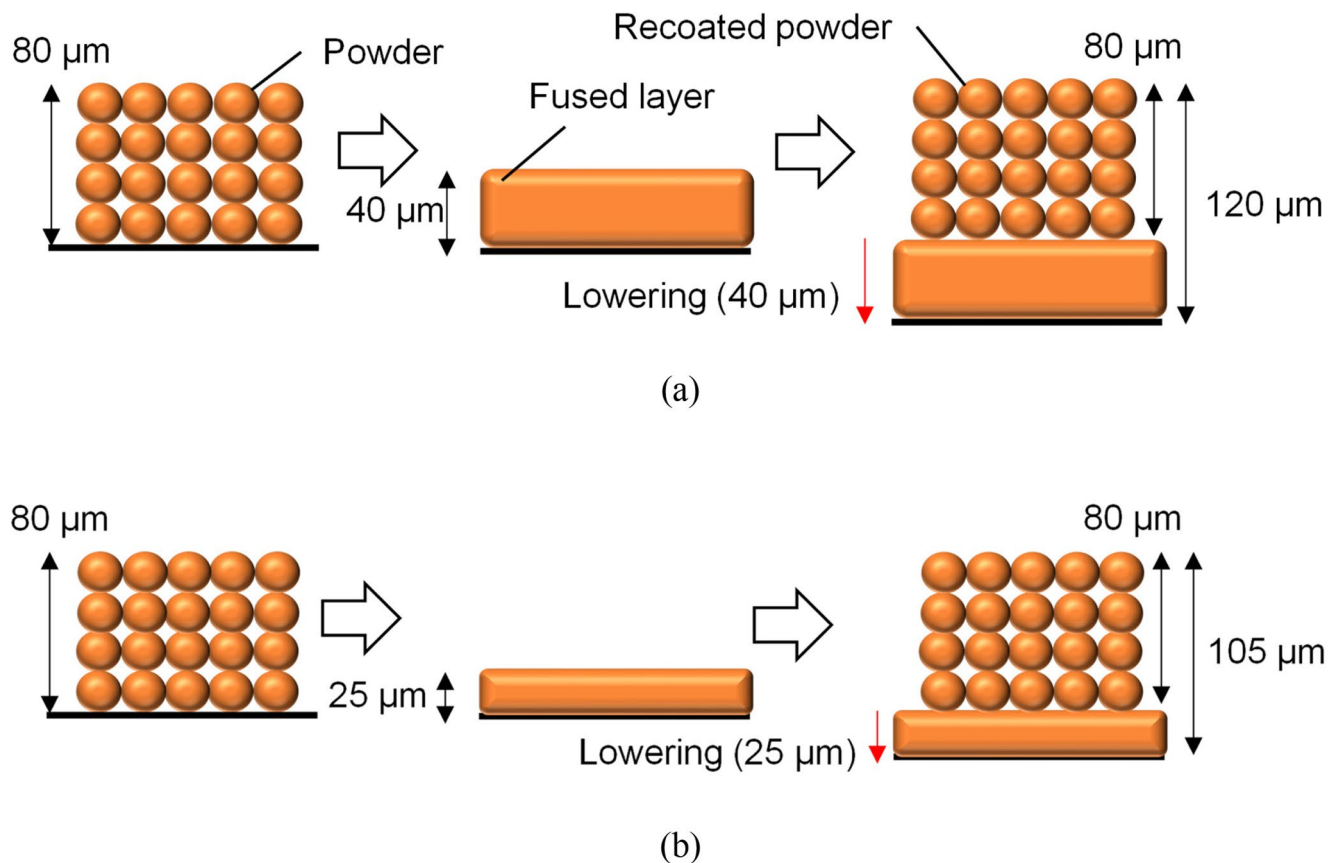


Fig. 10 Schematic of (a) conventional process and (b) improved process

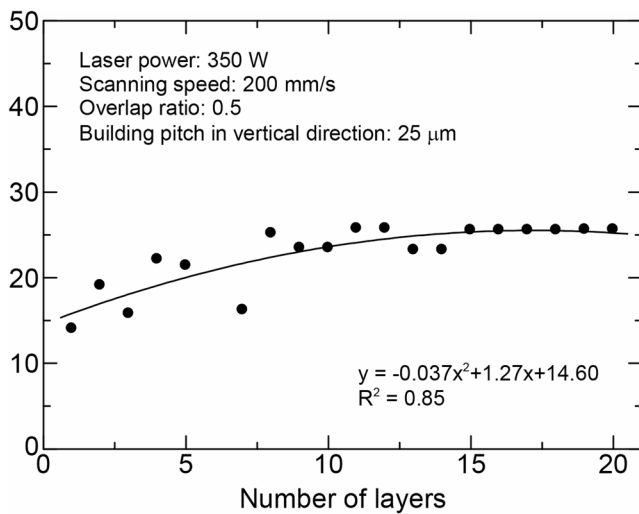


Fig. 11 Variation in built height per layer with the number of layers (building pitch in vertical direction: 25 μm)

In the improved process, the powder layer thickness for the first layer was set to 80 μm, and the platform lowering amount was adjusted to the predicted built height of 25 μm to ensure a constant powder layer thickness of 80 μm in each layer. The relationship between the measured built height and number of layers is shown in Fig. 11. The built height per layer was in the range of 14–26 μm. The built height per layer gradually increased up to the seventh layer, and its value was comparable to the predicted value of 25 μm above the eighth layer. When the built height is low, the heat diffusion into the pure copper substrate may affect the reduction in the built height. The built height became stable as the number of layers increased, which was attributed to the limitation of the heat diffusion region. The cross section of the structure built using the improved process is shown in Fig. 12. There was no significant change in the porosity depending on the

building height, and the porosity was decreased to 0.25% across the entire cross section. These results indicate that the thickness of the powder bed is also a key parameter that affects the porosity of the built structure, and it is essential to select an appropriate condition considering the difference in the density of the powder bed and built structure and the heat diffusion into the substrate.

In the PBF process of pure copper on a pure copper substrate using an infrared laser, Jadhav et al. [33] evaluated the relative density of copper parts fabricated using a fibre laser with a wavelength of 1080 nm and found that the relative density exceeded 98% at a VED of 740 J/mm³ or higher. In addition, Imai et al. [34] reported that a minimum VED of 667 J/mm³ was required to obtain a relative density of 98% or higher via PBF using a Yb-fibre laser. In contrast, for a stainless-steel substrate, Qu et al. [4] found that a relative density of 99.6% was attained at a VED of 667 J/mm³ in a PBF process using a fibre laser with a wavelength of 1070 nm. Their studies revealed that the stainless-steel and copper substrates required the same level of VED because the effect of the low thermal conductivity of steel is weakened as the number of layers increases, although its effect is noticeable in a single-track formation, as described in Sect. 3.1. In this study, the relative density of 99.75% of the copper part was achieved at a laser power of 350 W, scanning speed of 200 mm/s, hatching distance of 80 μm, and powder layer thickness of 80 μm, representing a VED of 270 J/mm³. These results indicate that the proposed PBF-LB/M system, which employs a wavelength synthesis method with a DDL, can produce copper parts on a pure copper substrate without preheating, and the high blue laser absorptivity of pure copper can reduce the VED by more than 50% compared with that of the infrared laser to obtain fully dense parts.

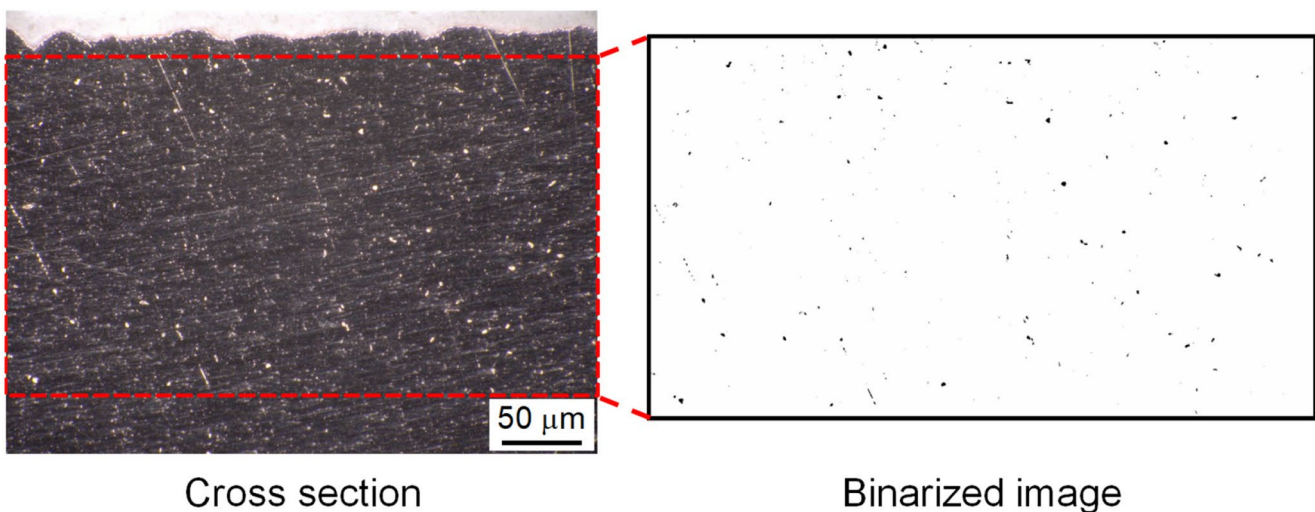


Fig. 12 Variation in porosity using the improved process

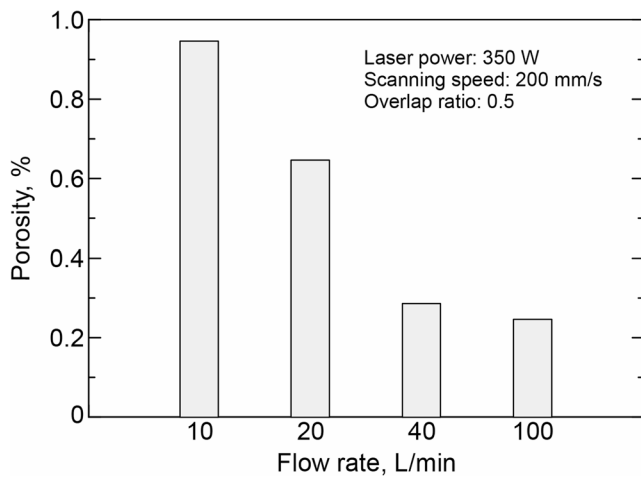
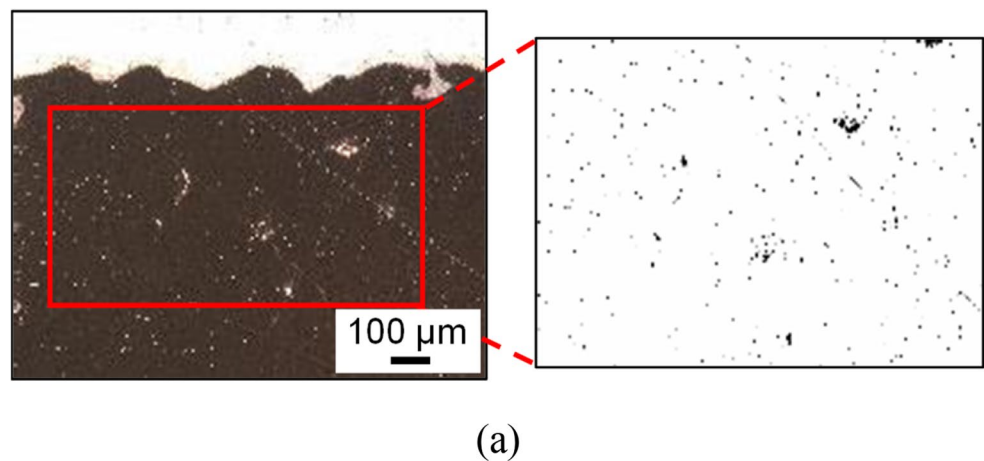


Fig. 13 Variation in porosity at different flow rates

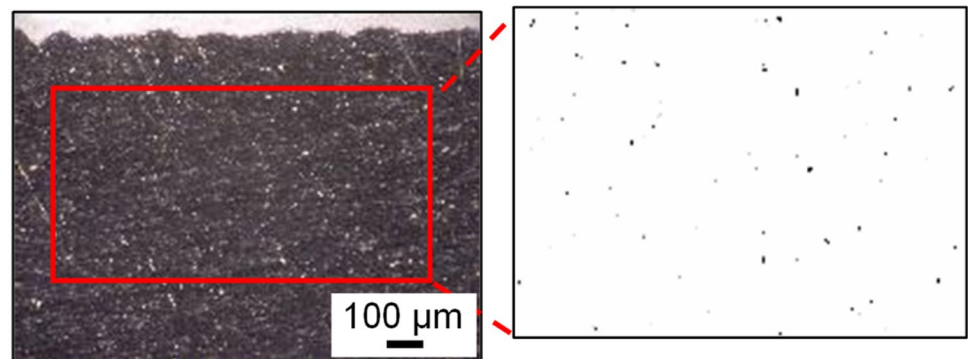
The variation in the porosity of the built structure fabricated at various flow rates of the atmospheric gas is shown in Fig. 13, and the cross sections of the built structures fabricated at flow rates of 10 and 100 L/min are compared in Fig. 14. The porosity decreased with an increase in the flow rate. A near-full density of the built structure was obtained at a flow rate of

40 L/min or higher. A high flow rate can improve the process stability because of the efficient removal of metal vapour and spatter particles from the building chamber, and stabilise the melt pool aspect owing to a reduction in the oxygen content in the shielding atmosphere. It has been reported that the intrusion of metal vapour and spatter particles causes deterioration of the beam quality [27]. The effect of metal vapour and spatter particles on the beam quality weakens at higher flow rates, increasing the process stability. In addition, a high oxygen content in the shielding gas is known to influence the temperature of the building area and Marangoni convection [35]. As strong Marangoni convection promotes spatter generation, a high gas flow rate may decrease the oxygen content in the building chamber, leading to more stable Marangoni convection and less spatter generation. Here, the efficient removal of metal vapour and spatter particles from the building chamber due to a high flow rate is considered to be the major factor contributing to the suppression of the porosity. Pure copper parts fabricated at a laser power of 350 W, scanning speed of 200 mm/s, and overlap ratio of 0.5 are demonstrated in Fig. 15. Parts with highly inclined overhangs and thin-walled sections were obtained without support structures.

Fig. 14 Comparison of the cross section of the built structure at a flow rate of (a) 10 L/min and (b) 100 L/min



(a)



(b)

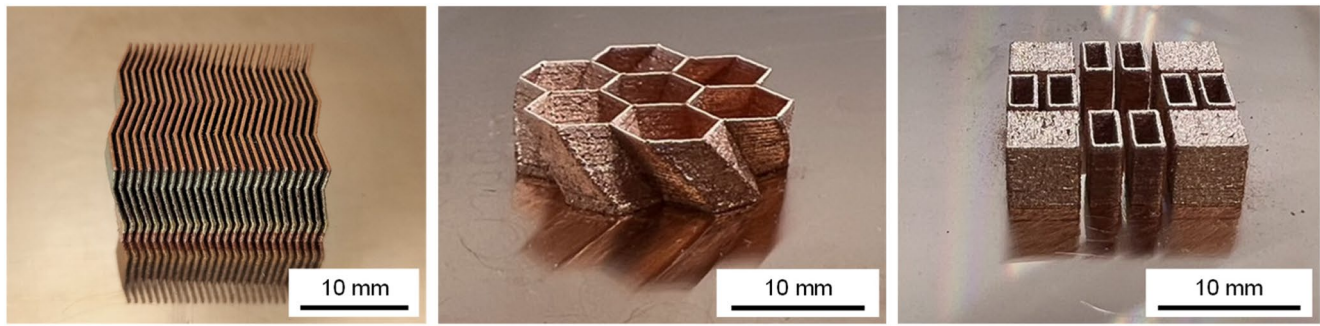


Fig. 15 Pure copper parts with highly inclined overhang and thin-walled section

4 Conclusions

This study investigated single- and multilayer formations in the PBF-LB/M process for pure copper parts using a blue laser with a wavelength of 440 nm. The following conclusions were drawn:

- 1) The proposed PBF-LB/M system, which employs a wavelength synthesis method with a DDL, can produce copper parts on a pure copper substrate without preheating.
- 2) A uniform geometry of a single track was obtained at a laser power of 300–350 W and scanning speed of 100–250 mm/s.
- 3) The arithmetic mean height P_a of a single layer decreased with an increase in the overlap ratio, and its value was the smallest, 8.1 μm , at an overlap ratio of 0.5. To achieve a structure without surface waviness, it is imperative to select a suitable overlap ratio based on the geometry of a single track.
- 4) At a powder layer thickness of 40 μm , the porosity of the built structure was 1.6%, and its value decreased to 0.25% after using an improved process that controlled the layer thickness to 25 μm in each layer.
- 5) The porosity of the built structure decreased with an increase in the flow rate of the atmospheric gas in the building chamber, and its value changed from 0.95% to 0.25% at a flow rate of 10–100 L/min.
- 6) A relative density of 99.75% of the built structure was achieved at a laser power of 350 W, scanning speed of 200 mm/s, hatching distance of 80 μm , and powder layer thickness of 80 μm , representing a VED of 270 J/mm³. The high blue laser absorptivity of pure copper reduced the VED by more than 50% compared with that of the infrared laser to obtain fully dense parts.

Author contribution Mitsugu Yamaguchi contributed to validation, data curation, writing of the original draft, and visualization. Kosei Kanie and Satoshi Abe were responsible for investigation and formal analysis. Kentaro Miyano contributed to conceptualization and provided resources. Naohiko Kobata and Yuki Somekawa were involved in investigation and methodology. Hiroshi Ohno and Tatsuaki Furumoto supervised all the research and analysis. All authors read and approved the final manuscript.

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Declarations

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Competing interests All authors declare that they have no competing.

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